

1. Company and Product Identification

- 1.1 Identification – Product Name: **Avista® 192**
- 1.2 Other means of identification: Organic and Inorganic Salts
 Synonym: Mixture, none
 Recommended Use of the: Reverse osmosis, microfiltration and ultrafiltration membrane cleaner.
- 1.3 Chemical and Restrictions on Use: Use only as directed on the label.
- 1.4 Name, Address, And Telephone Number of the Manufacturer, or other Responsible Party: **AVISTA TECHNOLOGIES**
 140 Bosstick Street
 San Marcos, CA 92069
 (760) 744-0536
 Competent Person email address: regulatory@avistatech.com
- 1.5 24 Hour Emergency No.: 1-800-424-9300 (United States)
 1-703 527-3887 (International Collect)



CERTIFIED BY NSF INTERNATIONAL TO NSF/ANSI 60 AS STANDARD DRINKING WATER TREATMENT CHEMICAL FOR USE IN REVERSE OSMOSIS SYSTEMS AT A MAXIMUM LEVEL OF 7 mg/l.

2. HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW: This product is an odorless, white to cream colored solid. This product can irritate contaminated skin, eyes, mucous membranes, and any other exposed tissues. This product is neither reactive nor flammable. Thermal decomposition of this product produces irritating vapors and toxic gases (e.g., carbon oxides, phosphorus oxides, and sodium oxides). Emergency responders must wear personal protective equipment (and have appropriate fire-extinguishing protection) suitable for the situation to which they are responding.

- Physical Hazards Summary: This product is a skin or eye irritant
- Potential Health Hazards Summary: Skin Corrosion/Irritation - Category 1B
 Serious Eye Damage Eye Irritation - Category 2A
- Potential Ecological Effects Summary: None
- 2.1 Classification of Product
- U.S. OSHA classification: Skin Corrosion/Irritation - Category 1B
 Serious Eye Damage Eye Irritation - Category 2A
- Classification as per EC 1272/2008 (CLP/GHS): Skin Corrosion/Irritation - Category 1B
 Serious Eye Damage Eye Irritation - Category 2A
- WHMIS classification: Skin Corrosion/Irritation - Category 1B
 Serious Eye Damage Eye Irritation - Category 2A

Hazardous Materials Information
System (HMIS) Rating

Health	2
Flammability	0
Physical Hazard	1
Protective Equipment	D

2.2 Label Elements OSHA/GHS

General Warnings	P101	If medical advice is needed, have product container or label at hand.
	P102	Keep out of reach of children.
	P103	Read label before use
	P403	Store in a well-ventilated place.
	P233	Keep container tightly closed
Signal Word	DANGER	
Hazard statements	H 272	Causes severe skin burns and eye damage.
	H314	Causes serious eye irritation
	H319	
Precautionary statements	P305	IF IN EYES, RINSE THOROUGHLY WITH RUNNING WATER
	P338	Remove contact lenses if present and easy to do. Continue rinsing.
	P280	Wear protective gloves/protective clothing/eye protection/face protection
	P264	Wash skin thoroughly after handling
	P271	Use only outdoors or in a well-ventilated area.
	P312	IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
	P302/P352	
	P337 + P313	If eye irritation persists: Get medical advice/attention.
	P404	Store in a closed container.
	P273	Avoid release to the environment.

Hazard pictograms - GHS



Hazard pictograms - WHMIS



2.3	Unclassified Hazards	None
2.4	Ingredients with unknown acute toxicity	None

3. COMPOSITION and INFORMATION ON INGREDIENTS

Chemical name CAS # EINECS #	% w/w	US OSHA	GHS/EU CLP	WHMIS
Silicate compound Proprietary Proprietary	70 - 70	Skin Corrosion/Irritation - Category 1B	Skin Corrosion/Irritation - Category 1B	Skin Corrosion/Irritation - Category 1B
Oxygenated inorganic salt Proprietary Proprietary	10 - 15	Acute Tox. 4 Eye Irrit. 2A Oxidizing solid Category 2	Acute Tox. 4 Eye Irrit. 2A Oxidizing solid Category 2	Acute Tox. 4 Eye Irrit. 2A Oxidizing solid Category 2
Polyphosphate Proprietary Proprietary	5 - 10	Acute Hazards to the aquatic environment - Category 3 Specific Target Organ Toxicity Single Exposure - Category 3 Skin Corrosion/Irritation - Category 1B Serious Eye Damage Eye Irritation - Category 1	Acute Hazards to the aquatic environment - Category 3 Specific Target Organ Toxicity Single Exposure - Category 3 Skin Corrosion/Irritation - Category 1B Serious Eye Damage Eye Irritation - Category 1	Acute Hazards to the aquatic environment - Category 3 Specific Target Organ Toxicity Single Exposure - Category 3 Skin Corrosion/Irritation - Category 1B Serious Eye Damage Eye Irritation - Category 1
Citrate compound Proprietary Proprietary	5 - 10	Irritant, Category 2	Irritant, Category 2	Irritant, Category 2
PRODUCT CLASSIFICATION	100	Skin Corrosion/Irritation - Category 1B Serious Eye Damage Eye Irritation - Category 2A	Skin Corrosion/Irritation - Category 1B Serious Eye Damage Eye Irritation - Category 2A	Skin Corrosion/Irritation - Category 1B Serious Eye Damage Eye Irritation - Category 2A

NE = Not Established. C = Ceiling Limit. See Section 16 for Definitions of Terms Used.

4. FIRST-AID MEASURES

4.1 Description of Necessary Measures

Skin exposure: If this product contaminates the skin, immediately begin decontamination with running water. Remove exposed or contaminated clothing, taking care not to contaminate eyes. Victim should seek immediate medical attention if any adverse exposure symptoms develop.

Eye exposure: If this product enters the eyes, open victim's eyes while under gently running water. Use sufficient force to open eyelids. Have victim "roll" eyes. Minimum flushing is for 15 minutes. Victim must seek medical attention.

Inhalation: If dusts of this product are inhaled, remove victim to fresh air. If necessary, use artificial respiration to support vital functions. Remove or cover gross contamination to avoid exposure to rescuers.

Ingestion: If this product is swallowed, CALL PHYSICIAN OR POISON CONTROL CENTER FOR MOST CURRENT INFORMATION. DO NOT INDUCE VOMITING. Have victim rinse mouth with water, if conscious. Never induce vomiting or give a diluent (e.g., water) to someone who is unconscious, having convulsions, or unable to swallow. If contaminated individual is convulsing, maintain an open airway and obtain immediate medical attention.

4.2 Most Important Symptoms/Effects:

Immediate: Inhalation exposure may cause coughing or sneezing. Symptoms of skin and eye contact may include burning, redness and irritation. Ingestion may cause stomach pains, cramps, and gastritis.

Delayed: Prolonged or repeated skin overexposure to this product may cause dermatitis (dry, red skin). Symptoms may include tingling, redness, and visible injury.

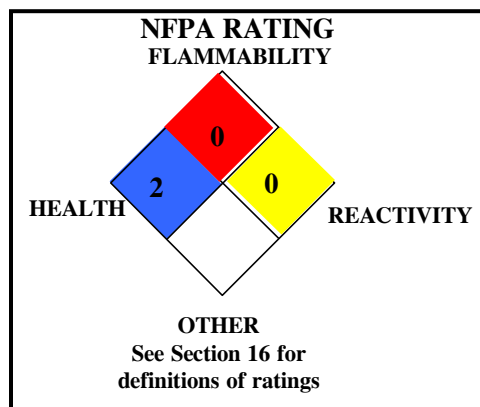
4.3 Indication of Immediate Medical Attention and Special Treatment Needed, If Necessary:

TARGET ORGANS: Acute: Skin, eyes. Chronic: Skin.

Victims of chemical exposure must be taken for medical attention if any adverse effects occur. Rescuers should be taken for medical attention if necessary. Take a copy of label and MSDS to physician or health professional with victim.

5. FIRE-FIGHTING MEASURES

Flammable properties Non-flammable oxidizing solid



Flash Point °C: Not applicable.

Autoignition Temperature °C: Not applicable.

Flammable Limits (in air by volume, %):

Upper: Not applicable.

Lower: Not applicable.

5.1 Suitable and Unsuitable
Extinguishing Media:

This material may contribute to the intensity of a fire. Use extinguishing material suitable to the surrounding fire.

Water spray	YES	Carbon dioxide	YES
Foam	YES	Dry chemical	YES
Halon	YES	Other	YES

5.2 Specific Hazards Arising from Chemical:

When involved in a fire, this material may decompose and produce irritating fumes and toxic gases (e.g., carbon monoxide, carbon dioxide, phosphorous oxides, and nitrogen oxides).

Explosion Sensitivity to Mechanical Impact: Not applicable.

Explosion Sensitivity to Static Discharge: Not applicable.

5.3 Special Protective Equipment and
Precautions for Fire-Fighters:

Incipient fire responders should wear eye protection. Structural firefighters must wear Self-Contained Breathing Apparatus and full protective equipment. Move containers from fire area if it can be done without risk to personnel. If possible, prevent runoff water from entering storm drains, bodies of water, or other environmentally sensitive areas.

6. ACCIDENTAL RELEASE MEASURES

- 6.1 Personal Precautions Uncontrolled releases should be responded to by trained personnel using pre-planned procedures. Proper protective equipment should be used. In case of a spill, clear the affected area and protect people.
- Protective equipment For small releases (< 20 kg), clean up spilled solid wearing gloves, goggles, faceshield, and suitable body protection. The minimum Personal Protective Equipment recommended for response to non-incident releases (more than 20 kg) should be Level C: triple-gloves (neoprene gloves and nitrile gloves over latex gloves), chemical resistant suit and boots, hard hat, and full-face respirator with HEPA filter.
- Emergency procedures Monitoring must indicate that exposure levels are below those provided in Section 8 (Exposure Controls-Personal Protection) and that oxygen levels are above 19.5% before anyone is permitted in the area without Self-Contained Breathing Apparatus.
- 6.2 Methods and Materials for Containment and Cleaning Up **KEEP AWAY FROM ORGANIC READILY COMBUSTIBLE MATERIALS.** Moisten to suppress dust. Shovel up solids into plastic container for recovery/disposal. Neutralize residue with sodium bicarbonate or other neutralizing agent for weak caustics. Decontaminate the area thoroughly. Test area with litmus paper to ensure neutralization. Place all spill residues in a suitable plastic container. Dispose of in accordance with applicable U.S. Federal, State, or local procedures, or appropriate local standards (see Section 13, Disposal Considerations).
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7. HANDLING and STORAGE

- 7.1 Precautions for Safe Handling All employees who handle this material should be trained to handle it safely. Open containers carefully on a stable surface. Empty containers may contain residual solid; therefore, empty containers should be handled with care.
- As with all chemicals, avoid getting this product ON YOU or IN YOU. Wash thoroughly after handling this product. Do not eat or drink while handling this material. Avoid generating dust of this product. Remove contaminated clothing immediately.
- During equipment maintenance follow practices indicated in Section 6 (Accidental Release Measures) to decontaminate equipment or clean-up small spills. Make certain that application equipment is locked and tagged-out safely if necessary. Collect all rinsates and dispose of according to applicable U.S. Federal, State, or local procedures or appropriate local standards.
- 7.2 Conditions for Safe Storage Store at temperatures less than 45°C (113°F). Store containers in a cool, dry location, away from direct sunlight, sources of intense heat, or where freezing is possible. Store away from incompatible materials. Material should be stored in secondary containers, or in a diked area, as appropriate. Storage and use areas should be covered with impervious materials. Keep container tightly closed when not in use. Store in original vented shipping container. If appropriate, post warning signs in storage and use areas. Inspect all incoming containers before storage, to ensure containers are properly labeled and not damaged.
- Incompatibilities Strong acids, oxidizers, caustics. It may react with metals to generate pressure.

8. EXPOSURE CONTROLS - PERSONAL PROTECTION

8.1 Control Parameters

CHEMICAL NAME	CAS #	% w/w	EXPOSURE LIMITS IN AIR					
			ACGIH-TLVs		OSHA-PELs		IDLH mg/m ³	OTHER mg/m ³
			TWA mg/m ³	STEL mg/m ³	TWA mg/m ³	STEL mg/m ³		
Silicate compound	Proprietary	70 - 75	NE	NE	NE	NE	NE	NE
Oxygenated inorganic salt	Proprietary	10 - 15	NE	NE	NE	NE	NE	NE
Polyphosphate	Proprietary	5 - 10	NE	NE	NE	NE	NE	NE
Citrate compound	Proprietary	5 - 10	NE	NE	NE	NE	NE	NE
			10 total; 3 respirable, mg/m ³ 50 mppcf total; 15 mppcf respirable					
Water and other components which are present in less than 1 percent concentration (0.1% concentration for potential carcinogens, reproductive toxins, respiratory tract sensitizers and mutagens).		Balance	None of the other components contribute significant additional hazards at the concentration present in this product. All pertinent hazard information has been provided in this document, per the requirements of the Federal Occupational Safety and Health Administration Standard (29 CFR 1910.1200), U.S. State equivalent Standards and Canadian Workplace Hazardous Materials Identification System Standards (CPR 4).					

8.2 Appropriate Engineering Controls.

Use with adequate ventilation to ensure exposure levels are maintained below the limits provided in this Section or as low as reasonably practical. Ensure eyewash/safety shower stations are available near areas where this product is used.

8.3 Personal Protective Equipment

Respiratory protection:

None needed under normal conditions of use. Use NIOSH approved respirators if ventilation is inadequate to control mists or vapor. If respiratory protection is needed, use only protection authorized in the U.S. Federal OSHA Standard (29 CFR 1910.134), applicable U.S. State regulations, or the applicable local standards. Oxygen levels below 19.5% are considered IDLH by OSHA. In such atmospheres, use of a full-face piece pressure/demand SCBA or a full-face piece, supplied air respirator with auxiliary self-contained air supply is required under OSHA's Respiratory Protection Standard (1910.134).

Eye protection:

Use approved safety goggles or safety glasses, as described in OSHA 29 CFR 1910.133. Splash goggles with a faceshield may be needed if splash hazards exist.

Hand protection:

Wear chemical impervious gloves (e.g., Solvex™, Neoprene).

Body protection:

If needed, use body protection appropriate for task (e.g., Tyvek suit, rubber apron) to protect from splashes and sprays.

9. PHYSICAL and CHEMICAL PROPERTIES

Appearance	This product is a white to cream colored powder.		
Odor	Slight to none	Odor Threshold	N/A
Melting Point °C	NE	pH (1% aqueous solution)	11.5 – 12.9
Initial Boiling Point °C	NE	Boiling Point Range °C	N/A
Flammability	Non-flammable	Evaporation Rate (water = 1)	N/A
Vapor Density (air = 1)	N/A	Vapor Pressure mm Hg @ 20°C:	N/A
Solubility (in water)	Soluble	Relative density (water = 1)	N/A
Viscosity	Flowing solid	Oil-Water Partition Coefficient	N/A
Decomposition Temperature	NE		
How to Detect This Substance (Warning Properties):	Litmus paper will turn blue when in contact with solutions of this product. Starch-iodide paper will turn dark.		

This product not an oxidizer per the U.S. DOT specifications based on similarity to related product. It does exhibit moderate oxidizer properties and must be handled as such.

10. STABILITY and REACTIVITY

10.1	Reactivity	Not considered reactive.
10.2	Chemical Stability	Stable
10.3	Possibility of hazardous reactions	Hazardous polymerization will not occur.
10.4	Conditions to avoid	Avoid mixing with incompatible materials.
10.5	Incompatible Materials	Strong acids, oxidizers, caustics. It may react with metals to generate pressure.
10.6	Hazardous Decomposition Products	Thermal decomposition of this product may generate carbon monoxide, carbon dioxide, phosphorous oxides and nitrogen oxides.

11. TOXICOLOGICAL INFORMATION

Toxicity data for hazardous ingredients	Oral LD ₅₀ mg/kg	Dermal LD ₅₀ mg/kg	Inhalation LD ₅₀ mg/kg
Silicate compound	LD ₅₀ (Oral, rat) 1153	N/A Skin irritation (24 hr) severe	N/A
Oxygenated inorganic salt	> 1034	> 2000	N/A
Polyphosphate	LD ₅₀ (oral, rat) > 7400 mg/kg LDLo (Intravenous-Rabbit, adult) 1580 mg/kg	LDLo (skin, rabbit) > 300 mg/kg	N/A
	Sex Chromosome Loss and Nondisjunction (Oral-Drosophila melanogaster) 11 pph Standard Draize Test (Skin-rabbit) > 300 mg/kg		
Citrate compound	LD ₅₀ (Oral-Rat) 3 g/kg LD ₅₀ (Oral-Mouse) 5040 mg/kg LD ₅₀ (Intraperitoneal-Rat) 883 mg/kg LD ₅₀ (Intraperitoneal-Mouse) 903 mg/kg LD ₅₀ (Subcutaneous-Rat) 5500 mg/kg LD ₅₀ (Subcutaneous-Mouse) 2700 mg/kg LD ₅₀ (Intraperitoneal-Mouse) LD50: 903 mg/kg LD ₅₀ (Intravenous-Rabbit, adult) 330 mg/kg LD ₅₀ (Intravenous-Mouse) 42 mg/kg LDLo (Oral-Rabbit, adult) 7000 mg/kg	LD ₅₀ (dermal, rabbit) > 2000 mg/kg	N/A
	Standard Draize Test (Skin-Rabbit, adult) 500 mg/24 hours: Moderate irritation effects Standard Draize Test (Eye-Rabbit, adult) 750 mg/24 hours: Severe irritation effects		

12. ECOLOGICAL INFORMATION

ALL WORK PRACTICES MUST BE AIMED AT ELIMINATING ENVIRONMENTAL CONTAMINATION.

12.1	Ecotoxicity	LC ₅₀ , mg/L	EC ₅₀ , mg/L
	Avista 192	342 <i>P. promelas</i> Fathead minnow	NE
	(Based on similar product)	41 <i>C. dubia</i> Water flea	
		250, NOEL, 96 hrs <i>P. promelas</i> Fathead minnow	
		31 NOEL, 48 hrs, <i>C. dubia</i> Water flea	
12.3	Bioaccumulative Potential	The components of this product are not expected to bioaccumulate. Significant releases could have an adverse impact on the pH of an aquatic system.	
12.4	Mobility in Soil	When spilled onto soil, this product will infiltrate downward, the rate being greater with lower concentration because of reduced viscosity.	
12.5	Other Adverse Ecological Effects	This product may be harmful to aquatic life <u>if large quantities</u> of it are released into an aquatic environment.	

13. DISPOSAL CONSIDERATIONS

Preparing Wastes of this Product for Disposal	Waste disposal must be in accordance with appropriate U.S. Federal, State, and local regulations or with local regulations. This product, if unaltered by the handling, may be disposed of by treatment at a permitted facility or as advised by your local waste regulatory authority.
Disposal of Contaminated Packaging	Cleaned containers can be recycled or disposed of as non-contaminated waste, if authorized by your local authorities. Dispose of containers as required by local regulations.
U.S. EPA Waste Number	D002, if this product becomes a waste

14. TRANSPORT INFORMATION

THIS MATERIAL IS HAZARDOUS AS DEFINED BY 49 CFR 172.101 BY THE U.S. DEPARTMENT OF TRANSPORTATION.

14.1	UN Number	UN3262
14.2	UN Proper Shipping Name	Corrosive solid, basic, inorganic, n.o.s. (Sodium Silicate, Sodium percarbonate)
14.3	Transport Hazard Class(es)	8 (Corrosive)
	Transport label(s) required	Corrosive Class 8
14.4	Packing Group	II
14.5	Marine Pollutant	Not applicable
	NA Emergency Response Guide Number (2012)	154
14.6	Transport in Bulk (Annex II of MARPOL 73/78 and IBC Code)	Not applicable
14.7	Special Transport Precautions	Not applicable
	National Motor Freight Classification	#70

International Air Transport Association

14.8	UN Number	UN3262
	UN Proper Shipping Name	Corrosive solid, basic, inorganic, n.o.s. (Sodium Silicate, Sodium percarbonate)
	Transport Hazard Class(es)	8 (Corrosive)
	Transport label(s) required	Corrosive Class 8
	Packing Group	II
	Packaging Instructions	863

14.9	UN Number	UN3262
	UN Proper Shipping Name	Corrosive solid, basic, inorganic, n.o.s. (Sodium Silicate, Sodium percarbonate)
	Transport Hazard Class(es)	8 (Corrosive)
	Transport label(s) required	Corrosive Class 8
	Packing Group	II
	Marine Pollutant	Not applicable
	NA Emergency Response Guide Number (2012)	154
	Transport in Bulk (Annex II of MARPOL 73/78 and IBC Code)	Not applicable

15. SAFETY, HEALTH and ENVIRONMENTAL REGULATIONS SPECIFIC FOR THE PRODUCT

PROGRAM	Polyphosphate	Oxygenated Inorganic Compound	Citrate Compound	Silicate Compound
US EPA PROGRAMS				
Clean Air Act Hazardous Air Pollutants	NO	NO	NO	NO
Safe Drinking Water Act	NO	NO	NO	NO
RCRA F, K, P, U or D-lists	NO	D001	NO	NO
SARA 302 RQ	NO	NO	NO	NO
SARA 302 TPQ	NO	NO	NO	NO
SARA 313 LISTED	NO	NO	NO	NO
SARA CHEMICAL CATEGORIES				
SARA 311/312 ACUTE	YES	YES	NO	YES
SARA 311/312 CHRONIC	YES	YES	NO	YES
SARA 311/312 FIRE	NO	NO	NO	NO
SARA 311/312 PRESSURE	NO	NO	NO	NO
SARA 311/312 REACTIVITY	NO	YES	NO	NO
EPA EXTREMELY HAZARDOUS SUBSTANCE	NO	NO	NO	NO
CALIFORNIA SAFE DRINKING WATER ACT (Proposition 65)				
This product does not contain any chemical listed on the California Safe Drinking Water Act list (Proposition 65)				
US OSHA PROGRAMS				
PEL	NO	NO	NO	YES
PSM	NO	NO	NO	NO
CHEMICAL SECURITY PROGRAMS				
DHS CFATS	NO	NO	NO	NO
CHEMICAL WEAPONS CONVENTION				
	NO	NO	NO	NO
US DRUG ENFORCEMENT ADMINISTRATION				
DEA Controlled Substances	NO	NO	NO	NO
CHEMICAL INVENTORY PROGRAMS				
WHMIS	D2B	E	D2B	D2B
DSL	YES	YES	YES	YES

REACH Pre-registered List	YES	YES	YES	YES
TSCA	YES	YES	YES	YES
European Inventory of Existing Commercial Chemical Substances (EINECS)	YES	YES	YES	YES
EU No-Longer Polymers List (NLP)	N/A	N/A	N/A	N/A
Philippines	YES	YES	YES	YES
Japan	YES	YES	YES	YES
Australia	YES	YES	YES	YES
Korea	YES	YES	YES	YES
China	YES	YES	YES	YES
New Zealand Inventory of Chemicals	YES	YES	YES	YES

16. OTHER INFORMATION

16.1	Original Preparation	26 March 2018
16.2	Revision History	01 Nov 2018 Content Corrections, 01 Jan 2020 Content Corrections, 20 April 2020 Content Corrections
16.3	Prepared by	ADVANCED CHEMICAL SAFETY, Inc. PO Box 152329 San Diego, CA 92195 (858)-874-5577
16.4	Date of Printing	April 23, 2020

DEFINITIONS OF TERMS

16.5	A large number of abbreviations and acronyms appear on a MSDS. Some of these which are commonly used include the following:	
	Section 2	GHS: Global Harmonization System OSHA: U.S. Occupational Safety and Health Administration. CLP: Classification and Packaging WHMIS: Workplace Hazardous Materials Information System STOT: Specific Target Organ Toxicity
	Section 3	CAS #: Chemical Abstract Service index number EINECS #: European Chemical Substances Information System index number
	Section 5	NFPA: Nation Fire Protection Association Health Hazard: 0 (material that on exposure under fire conditions would offer no hazard beyond that of ordinary combustible materials); 1 (materials that on exposure under fire conditions could cause irritation or minor residual injury); 2 (materials that on intense or continued exposure under fire conditions could cause temporary incapacitation or possible residual injury); 3 (materials that can on short exposure could cause serious temporary or residual injury); 4 (materials that under very short exposure could cause death or major residual injury). Flammability Hazard Reactivity Hazard: Refer to definitions for "Hazardous Materials Identification System". Flash Point: Minimum temperature at which a liquid gives off sufficient vapors to form an ignitable mixture with air. Autoignition Temperature: The minimum temperature required to initiate combustion in air with no other source of ignition. LEL: The lowest percent of vapor in air, by volume, that will explode or ignite in the presence of an ignition source. UEL: The highest percent of vapor in air, by volume, that will explode or ignite in the presence of an ignition source.
	Section 8	ACGIH - American Conference of Governmental Industrial Hygienists, a professional association which establishes exposure limits. TLV - Threshold Limit Value - an airborne concentration of a substance which represents conditions under which it is generally believed that nearly all workers may be repeatedly exposed without adverse effect. The duration must be considered, including the 8-hour Time Weighted Average (TWA), the 15-minute Short Term Exposure Limit, and the instantaneous Ceiling Level (C). Skin absorption effects must also be considered PEL - Permissible Exposure Limit - This exposure value means exactly the same as a TLV, except that it is enforceable by OSHA. The OSHA Permissible Exposure Limits are based in the 1989 PELs and the June, 1993 Air Contaminants Rule (Federal Register : 58: 35338-35351 and 58: 40191). Both the current PELs and the vacated PELs are indicated. The phrase, "Vacated 1989 PEL," is placed next to the PEL which was vacated by Court Order. IDLH - Immediately Dangerous to Life and Health - This level represents a concentration from which one can escape within 30-minutes without suffering escape-preventing or permanent injury. The DFG - MAK is the Republic of Germany's Maximum Exposure Level, similar to the U.S. PEL. NIOSH is the National Institute of Occupational Safety and Health, which is the research arm of the U.S. Occupational Safety and Health Administration (OSHA). NIOSH issues exposure guidelines called Recommended Exposure Levels (RELs) . When no exposure guidelines are established, an entry of NE (Not Established) is made for reference.
	Section 11	LD₅₀ : Lethal Dose (solids & liquids) which kills 50% of the exposed animals; LC₅₀ : Lethal Concentration (gases) which kills 50% of the exposed animals; ppm: Concentration expressed in parts of material per million parts of air or water; mg/m³ : Concentration expressed in weight of substance per volume of air; mg/kg: Quantity of material, by weight, administered to a test subject, based on their body weight in kg IARC - the International Agency for Research on Cancer; NTP - the National Toxicology Program, RTECS - the Registry of Toxic Effects of Chemical Substances, OSHA and CAL/OSHA. IARC and NTP rate chemicals on a scale of decreasing potential to cause human cancer with rankings from 1 to 4. Subrankings (2A, 2B, etc.) are also used. TDLo, the lowest dose to cause a symptom and TCLo the lowest concentration to cause a symptom; TDo, LDLo, and LDo, or TC, TCo, LCLo, and LCo, the lowest dose (or concentration) to cause lethal or toxic effects. BEI - Biological Exposure Indices, represent the levels of determinants which are most likely to be observed in specimens collected from a healthy worker who has been exposed to chemicals to the same extent as a worker with inhalation exposure to the TLV.
	Section 12	LC₅₀: The lowest concentration in water which kills 50% of the test subjects. EC₅₀: The Effect Concentration in water at which 50% of the test species if affected.
	Section 13	US EPA Hazardous Waste Codes: refer to 40 CFR 261.20
	Section 14	DOT: US Department of Transportation IATA: International Air Transport Association IMO: International Maritime Organization MARPOL: International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978 IBC Code : Merchant Shipping Code
	Section 15	RCRA: US Resource Conservation and Recovery Act SARA: US Superfund Amendments and Reauthorization Act PSM: US OSHA Process Safety Management CFATS: US Department of Homeland Security Chemical Facility Anti-terrorism Standard DSL: Canadian Domestic Substances List NDSL: Canadian Non-Domestic Substances List REACH: European Registration, Evaluation, Authorization and Restriction of Chemicals list TSCA: US Toxic Substances Control Act